AWS Task 2

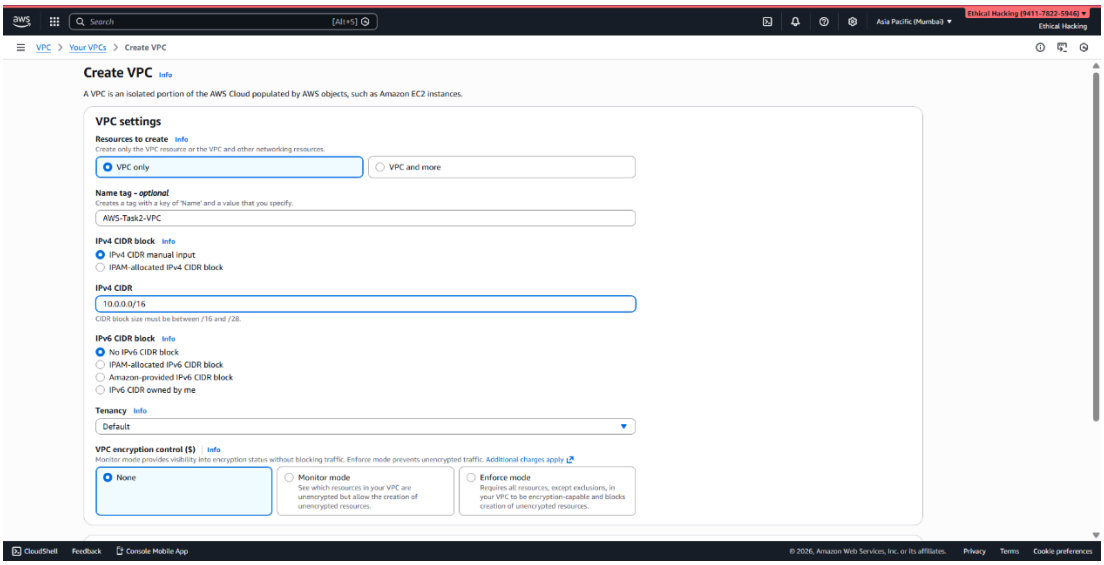
Task Description:

Set up a VPC with an Internet gateway, create a public subnet with 256 IP addresses, a private subnet with 256 IP addresses, make a route table connecting the Internet gateway and the subnets, and launch a Linux EC2 instance by using the above VPC and public subnet.

Solution:

Step 1: Creating a VPC

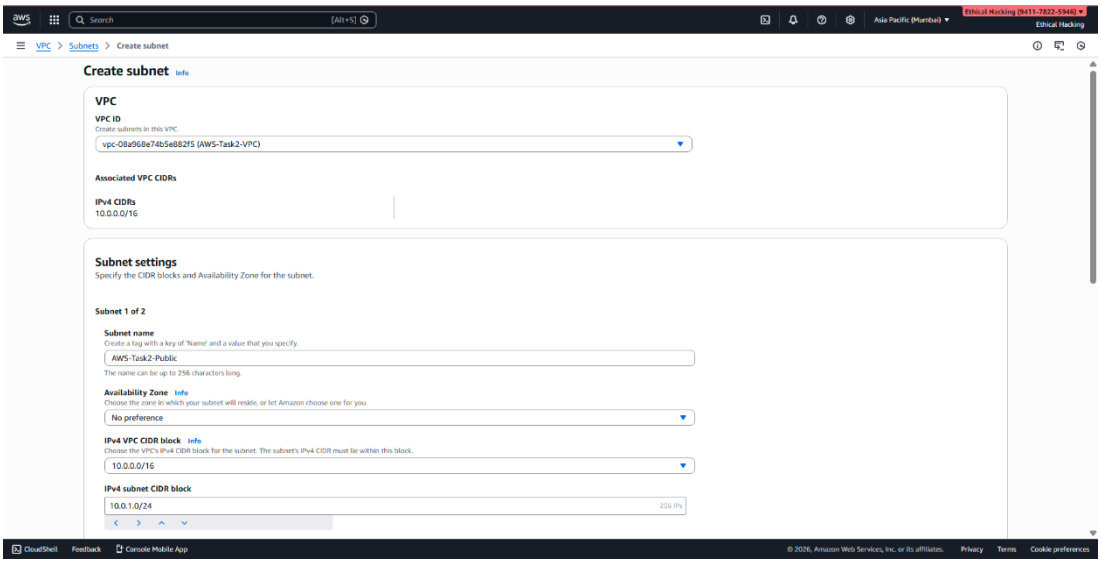
Using the IP range 10.0.0.0/16 for this VPC.

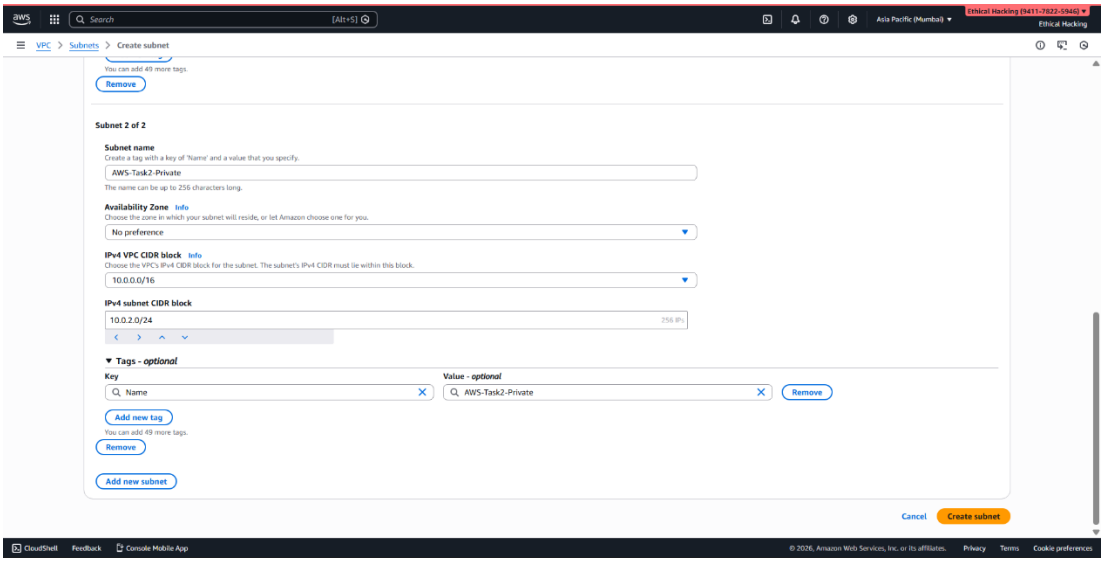


Step 2: Creating Subnets

For the Public IP addresses which we need - 256 IP, 2^8 = 256, I am using 10.0.1.0/24

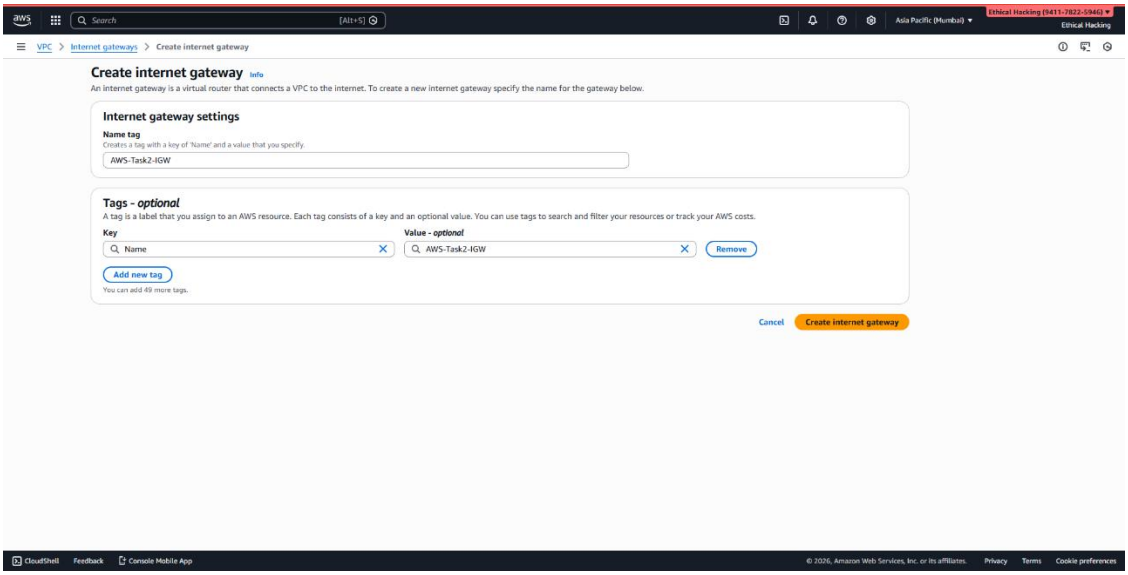
calculated this 24 using total 32 bits – the 8 host bits = 24 Network bits.



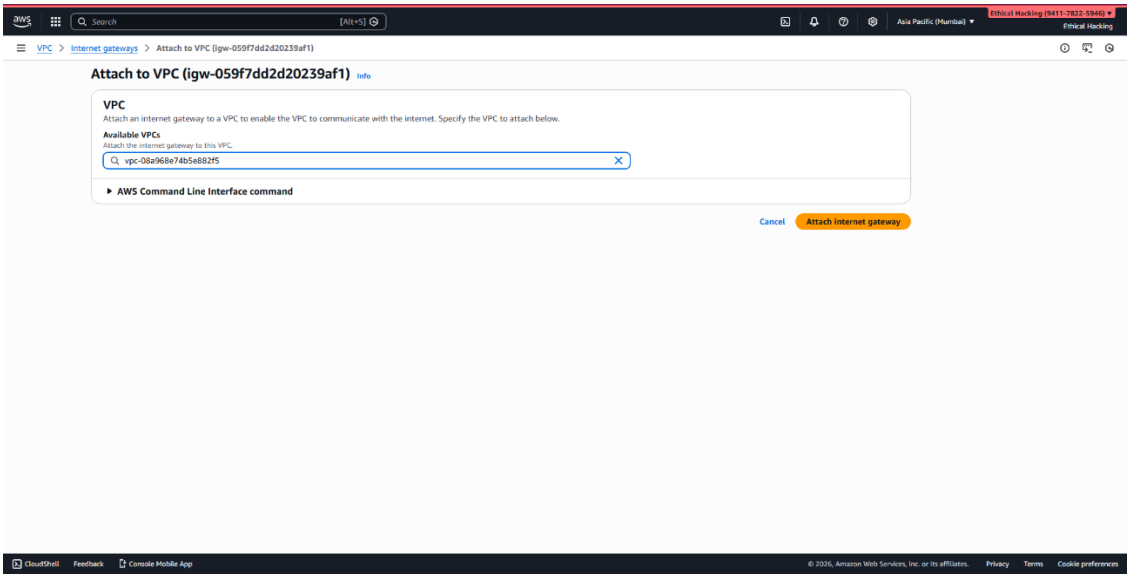


Step 3: Creating Internet Gateway

Gateway is needed so that the Public Subnet will be able to access the internet.

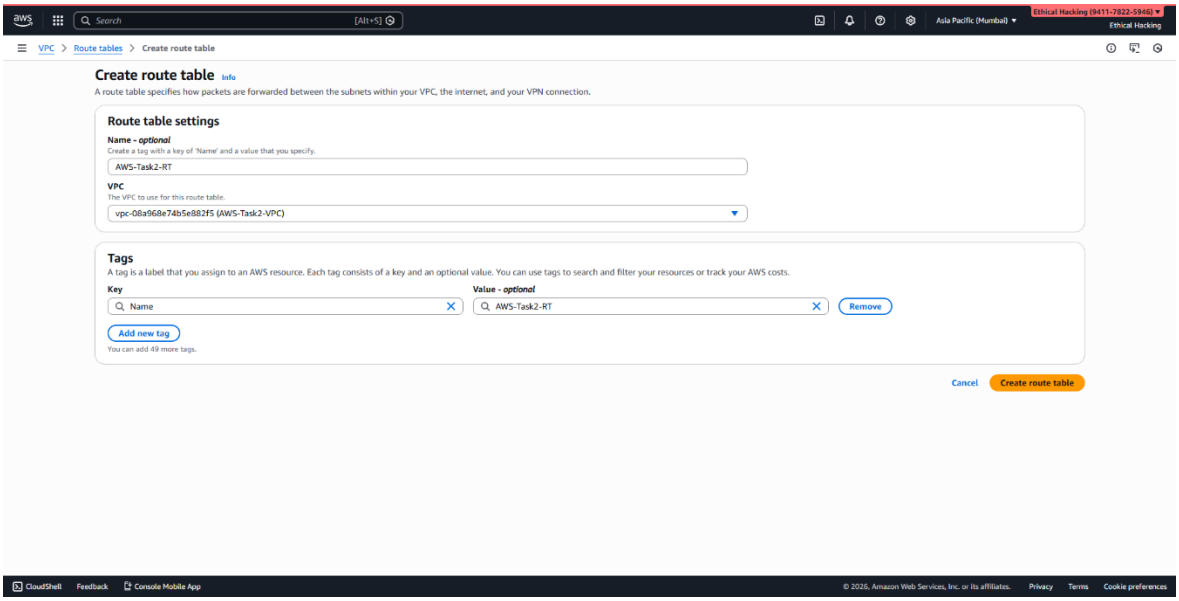


Step 4: Attaching the created Internet Gateway to the VPC



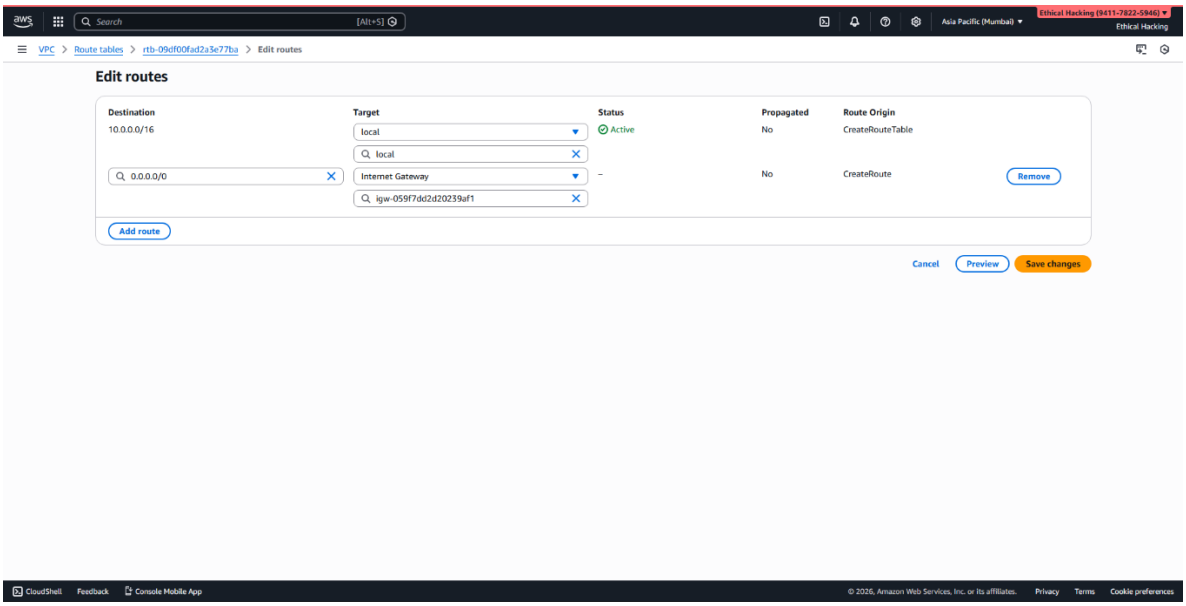
Step 5: Creating Route Table

Route table is necessary for the network traffic to reach it's proper destination, as Route table will be having all the route details.



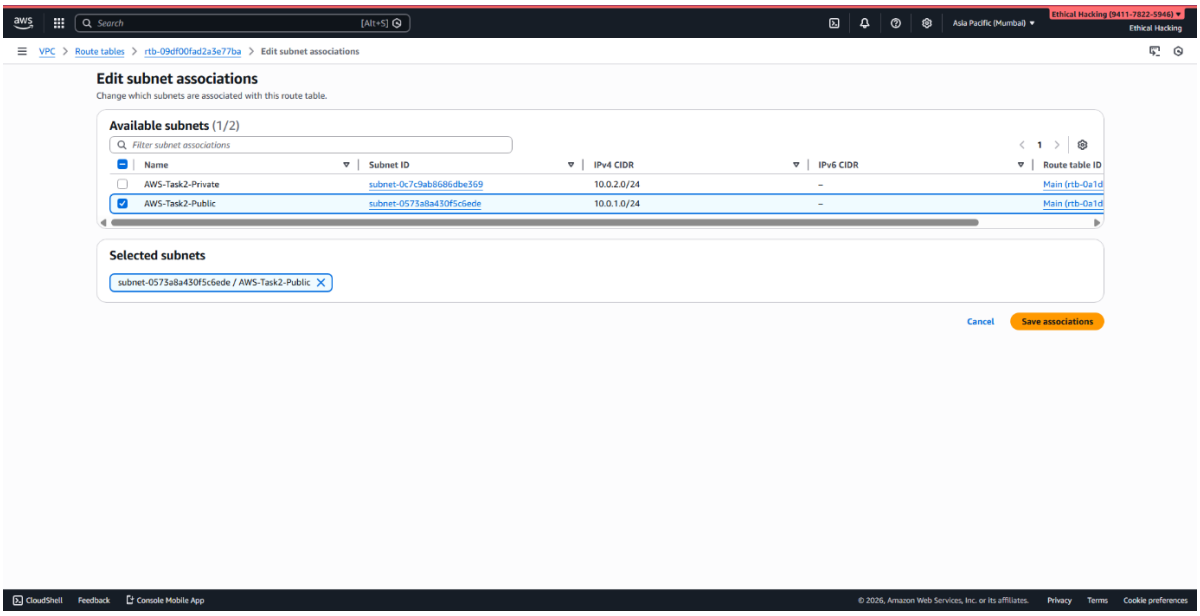
Step 6: Adding Internet Gateway to the Route Table

Adding the Internet Gateway to the Route Table so that the Route table can show the way for Internet to the public subnet IP's.

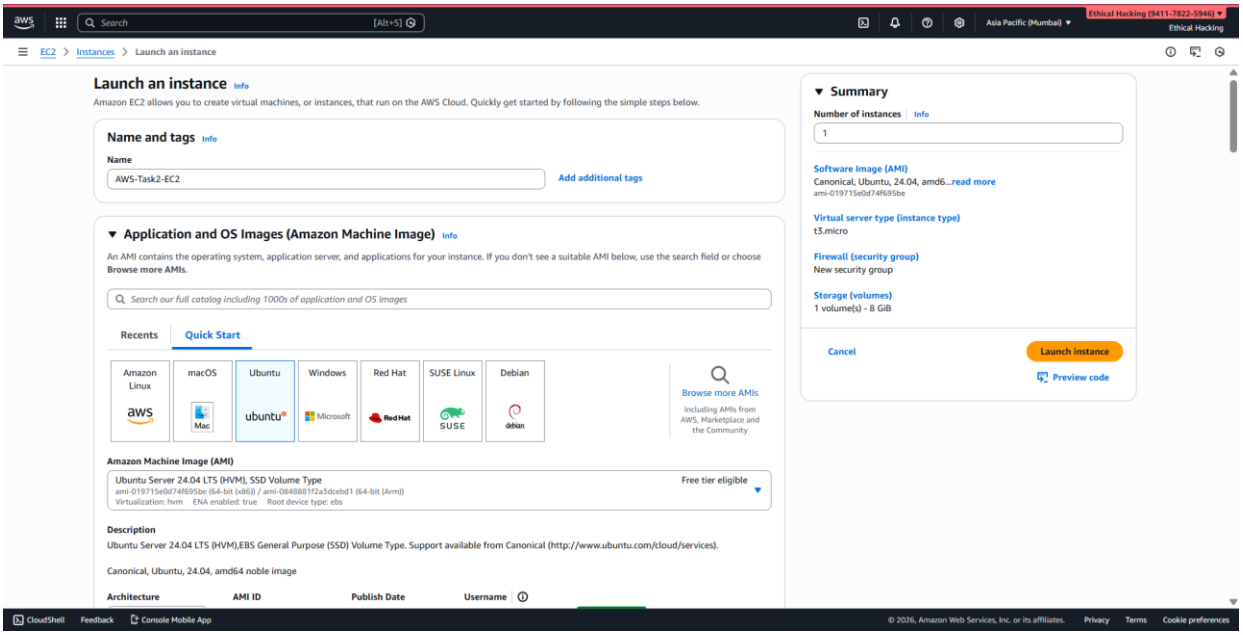


Step 7: Adding the Public IP subnet to the Route Table

Adding Public IP Subnet to the Route Table, so that when the Public IP need to connect somewhere, it can check the Route table for the exact path.



Step 8: Creating an EC2 Instance using the new VPC and Subnets created now



Network settings Info

VPC - *required* | Info

vpc-08a968e74b5e882f5 (AWS-Task2-VPC)
10.0.0.0/16

Subnet | Info

subnet-0573a8a430f5c6ede AWS-Task2-Public
VPC: vpc-08a968e74b5e882f5 Owner: 941178225946 Availability Zone: ap-south-1b (aps1-az3)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24)

Create new subnet

Auto-assign public IP | Info

Enable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (●) Create security group
- ( ) Select existing security group

Security group name - *required*

launch-wizard-3

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* | Info

launch-wizard-3 created 2026-02-07T19:25:15.587Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0) — Remove

Type	Protocol	Port range
ssh	TCP	22

Source type	Source	Description - *optional*
Anywhere	Add CIDR, prefix list or security group 0.0.0.0/0	e.g. SSH for admin desktop

Security group rule 2 (TCP, 80, 0.0.0.0/0) — Remove

Type	Protocol	Port range
HTTP	TCP	80

Source type	Source	Description - *optional*
Anywhere	Add CIDR, prefix list or security group 0.0.0.0/0	e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

User data - *optional* | Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
# Exit immediately if a command exits with a non-zero status
set -e

# Redirect all output to a log file so you can debug if things fail
# Access logs via: cat /var/log/user-data.log
exec > >(tee /var/log/user-data.log|logger -t user-data -s 2>/dev/console) 2>&1

echo "Starting deployment script..."

# ----------------------------
# 1) Update and install Apache
# ----------------------------
# Use -qq for "quite" mode and DEBIAN_FRONTEND to prevent interactive prompts
export DEBIAN_FRONTEND=noninteractive
```

☐ User data has already been base64 encoded

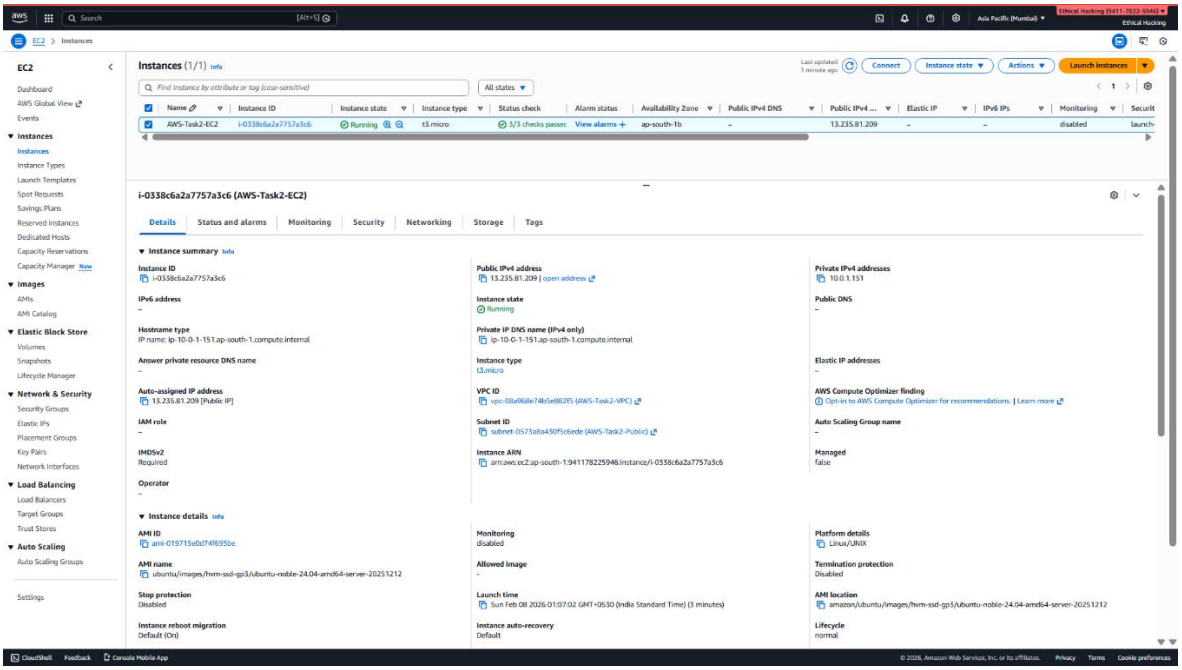
Code:

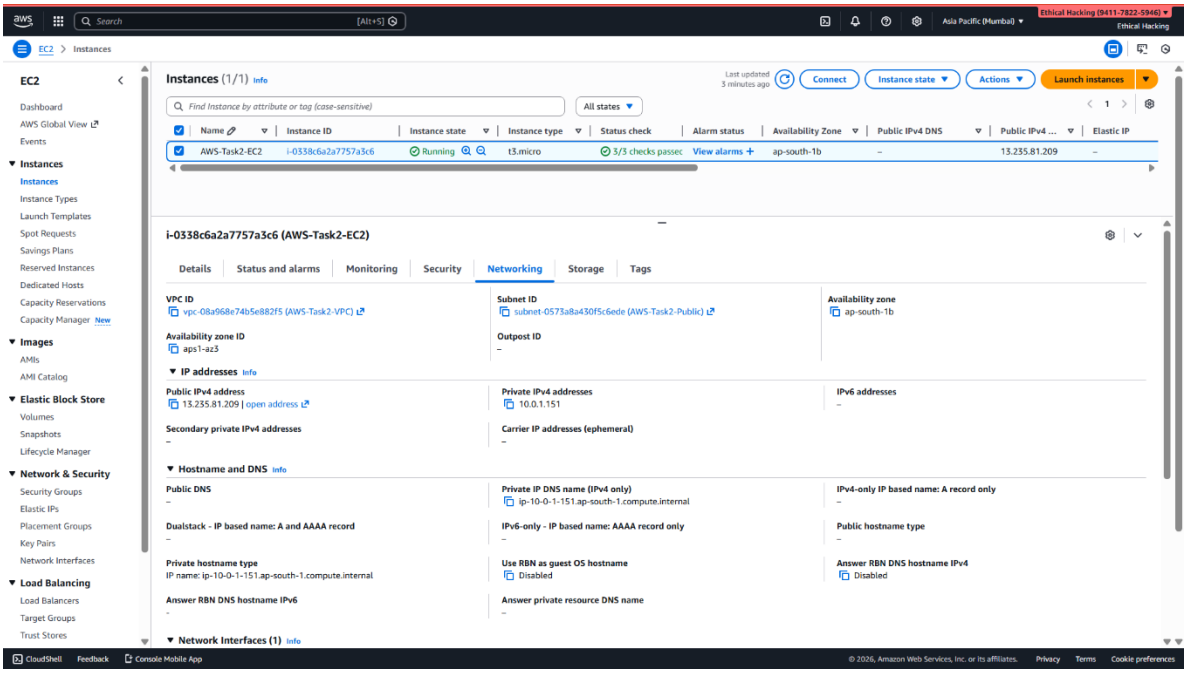
```
#!/bin/bash
set -e
exec > >(tee /var/log/user-data.log|logger -t user-data -s 2>/dev/console) 2>&1
export DEBIAN_FRONTEND=noninteractive
apt-get update -y
apt-get install -y apache2
systemctl enable apache2
systemctl start apache2
if command -v ufw >/dev/null 2>&1 && ufw status | grep -q "active"; then
  ufw allow 'Apache'
fi
cat <<'EOF' > /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Random Background</title>
  <style>
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      min-height: 100vh;
      display: grid;
      place-items: center;
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: #000000;
      transition: background-color 0.4s ease;
    }
    .container {
      background: rgba(255, 255, 255, 0.1);
      backdrop-filter: blur(10px);
      -webkit-backdrop-filter: blur(10px);
      border: 1px solid rgba(255, 255, 255, 0.2);
```

```
        padding: 40px;
        border-radius: 20px;
        text-align: center;
        box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37);
      }
      .text {
        font-size: 32px;
        font-weight: bold;
        color: #ffffff;
        margin-bottom: 24px;
        text-transform: uppercase;
        text-shadow: 2px 2px 4px rgba(0,0,0,0.5);
      }
      button {
        padding: 12px 24px;
        font-size: 16px;
        border: none;
        border-radius: 50px;
        cursor: pointer;
        font-weight: 600;
        background: #ffffff;
        color: #000000;
        box-shadow: 0 4px 15px rgba(0,0,0,0.2);
        transition: all 0.2s ease;
      }
      button:hover { transform: translateY(-2px); box-shadow: 0 6px 20px rgba(0,0,0,0.3); }
      button:active { transform: translateY(0); }
    </style>
  </head>
  <body>
    <div class="container">
      <div class="text">#000000</div>
      <button type="button" onclick="color()">Change Color</button>
    </div>
```

```
  <script>
    function color() {
      const randomColor = Math.floor(Math.random() * 16777215).toString(16).padStart(6, '0');
      const hex = "#" + randomColor;
      document.body.style.backgroundColor = hex;
      document.querySelector(".text").innerText = hex;
    }
  </script>
</body>
</html>
EOF
chown -R www-data:www-data /var/www/html
chmod -R 755 /var/www/html
rm -f /var/www/html/index.nginx-debian.html
systemctl restart apache2
```

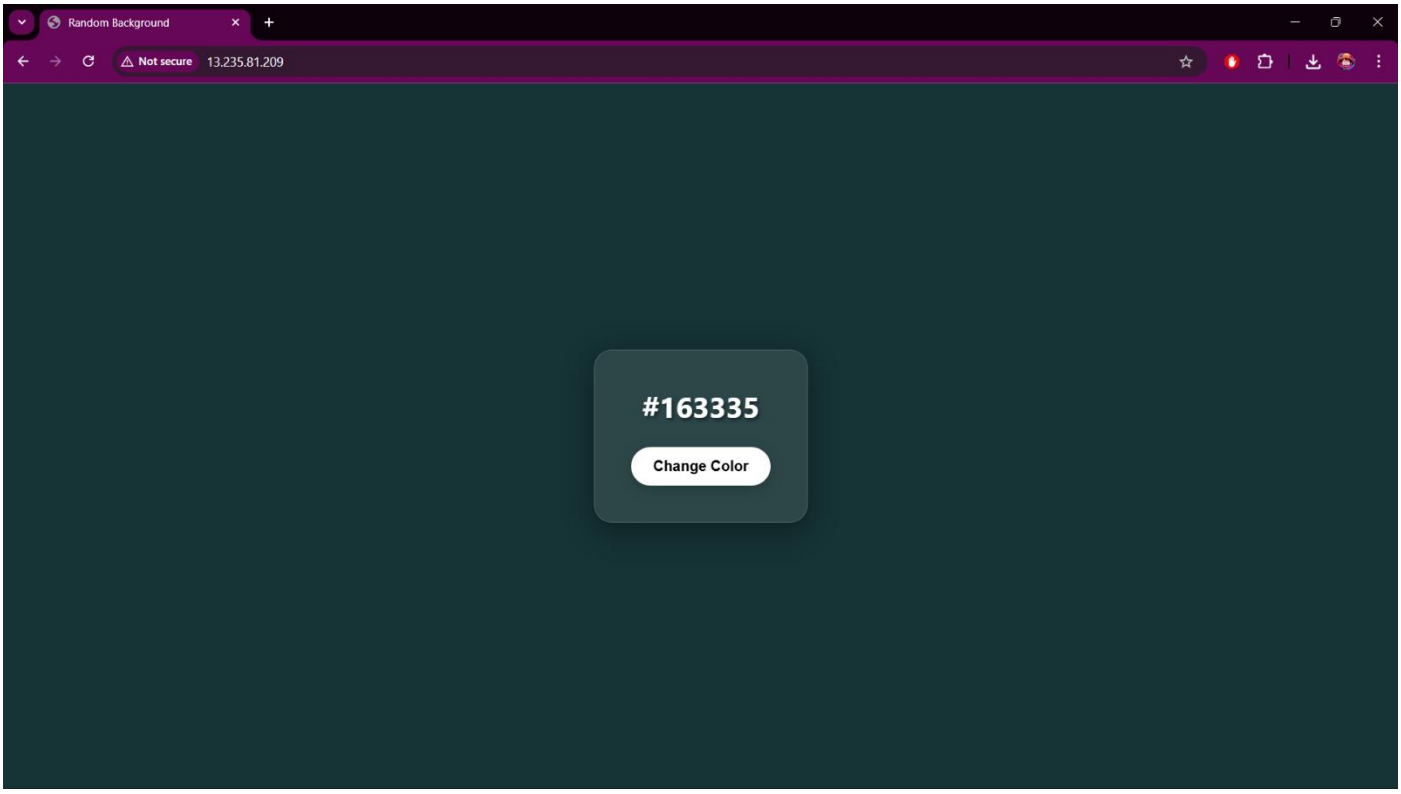
Final Output:





IP: 13.235.81.209

Result:


Random Background
Not secure
13.235.81.209
#163335
Change Color